Assignment 3: Transformer is All You Need

Tiny Shakespeare Next-Token Model (From-Scratch Transformer)

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Code: https://github.com/zz3231/COMS4995-Applied-Machine-Learning-HW/tree/main/HW3

1 Introduction

Causal language modeling on a fixed corpus is a clean testbed for the Transformer: every position is a supervised classification problem over the vocabulary, and the only structural prior is the ordering of tokens. Tiny Shakespeare is small enough to train a hand-written model on one GPU, yet large enough that nontrivial syntactic and stylistic regularities appear in the data. This work implements a compact decoder-only Transformer in PyTorch—Byte-Pair Encoding capped at 500 types (trained on the training split only), two layers, hidden width 128, eight heads, RMSNorm, sinusoidal positional encodings, and causal self-attention—and reports validation perplexity $\mathrm{PPL} = \exp(\mathrm{CE}_{\mathrm{val}})$ as the primary score.

The best model in the main configuration reaches PPL = 41.33 after 25 epochs with AdamW at learning rate 3×10^{-4}. The rest of the paper connects that aggregate number to internal behavior. Section 2 summarizes data and training protocol. Section 3 walks through training curves, attention maps at convergence, the evolution of one head across checkpoints, a layer-1 entropy summary across heads, a positional-encoding ablation, a shorter-context run, a small learning-rate study, and a brief generation sample. Section 4 pulls the threads together and notes limitations.

2 Methods

The raw file is split 80%/20% by character index so validation text never appears in the training string. A ByteLevel BPE tokenizer is fit on `train_text` with `vocab_size` = 500 and `min_frequency` = 2. Encoded sequences are cut into overlapping windows of length 65 with stride 32, yielding about 1.28×10^4 training windows and 64 next-token targets per window.

Each Transformer block uses pre-layer normalization and residual connections around causal multi-head attention and a two-layer feed-forward network. Attention scores use scaled dot products with a lower-triangular mask ($-\infty$ above the diagonal) before softmax. Sinusoidal positional encodings are added to token embeddings after scaling embeddings by $\sqrt{d_{\mathrm{model}}}$. The stack ends with RMSNorm and a linear map to logits. Parameter count is on the order of 10^6.

Optimization uses AdamW with learning rate 3×10^{-4}, weight decay 0.01, batch size 32, dropout 0.1, and gradient clipping at global norm 1.0. Random seeds are fixed for reproducibility; validation uses `torch.no_grad()`. For analysis, checkpoints are saved after epochs 1, 5, 15, and 25. A second model omits sinusoidal PE but is otherwise identical; it trains 15 epochs and is compared epoch-by-epoch to the first 15 epochs of the main run. A third model uses block size 33 and stride 16 for 12 epochs. For layer 1, mean row entropy of attention weights is computed on a fixed validation prefix. Finally, three fresh initializations each run 8 epochs at learning rates 10^{-4}, 3×10^{-4}, and 10^{-3} with the same optimizer settings otherwise. Numeric traces are logged in `hw3_exp_*.txt`.

3 Results

3.1 Training dynamics

Figure 1 plots training and validation cross-entropy and validation perplexity. Training loss falls from roughly 5.06 to 3.38 over 25 epochs; validation loss ends at 3.72, which corresponds to PPL = 41.33. The two curves start close, then the training loss pulls away after about epoch 5. That widening gap is the usual signature of a model that is beginning to memorize idiosyncrasies of the training windows while still improving on held-out

text—here validation perplexity decreases throughout, so there is no evidence of a damaging overfit regime within this horizon.

The perplexity curve makes the same story easier to read in "effective branching factor" units. Most of the improvement happens before epoch 5: PPL drops from 83.5 to about 48.6. Later epochs chip away at error more slowly, which suggests that coarse structure (punctuation, frequent subwords, line-level habits) is learned first and rare continuations dominate the remaining loss.

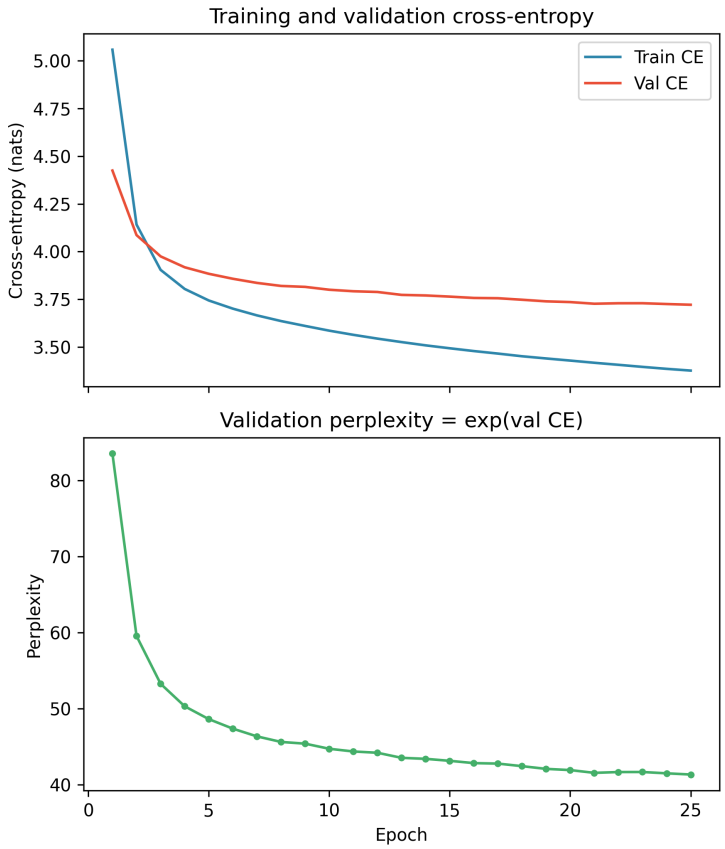

Figure 1: Training and validation cross-entropy (top) and validation perplexity (bottom).

Table 1: Summary of reported validation perplexities.

Item	Value
Main run (25 ep)	val CE 3.72; PPL 41.33
Checkpoints (L1H2 evolution)	ep1 83.50; ep5 48.61; ep15 43.13; ep25 41.33
No sinusoidal PE (15 ep)	final PPL 43.58 (main at ep15: 43.13)
Short context, block 33 (12 ep)	final PPL 40.19
LR probe (8 ep each)	final PPL: 10^{-4} 55.25; 3×10^{-4} 45.99; 10^{-3} 41.39

3.2 Attention at convergence

Figure 2 compares layer 0, head 0 and layer 1, head 2 on the first sequence of a validation batch (ByteLevel subword labels on both axes). The decoded prefix reads like early modern dialogue: roughly "you are, That is, a woman; if you be more, you're none; if you be." Every panel respects the causal mask: each row has support only on columns to the left of the diagonal, so no query looks past its own position.

The layer-0 map reads as a local mixer. Early positions concentrate probability on themselves and on immediate neighbors, producing a visible diagonal band. There is also a pronounced off-diagonal link from the piece "om" (inside "woman") back to "are", which is consistent with learning subject–predicate structure in this corpus. Commas and semicolons participate in vertical and horizontal structure, as if the head uses punctuation to segment clauses. For queries late in the prefix, weights spread over many keys at moderate magnitude rather than collapsing to a single predecessor; that pattern fits the idea that the first block aggregates a broad context vector for the second block to refine.

Layer 1, head 2 is noticeably sparser. The row for “is” peaks on “ou” (part of “you”). The row for “more” peaks on an earlier occurrence of “you”. The row for “re” in “you’re” also reaches back toward “you”. Several delimiter tokens attract mass from later rows. Taken together, the two heads illustrate a division of labor that shows up even in a two-layer stack: one head spreads mass locally and through punctuation, while another acts more like a pointer that tracks a repeated pronoun and clause boundaries.

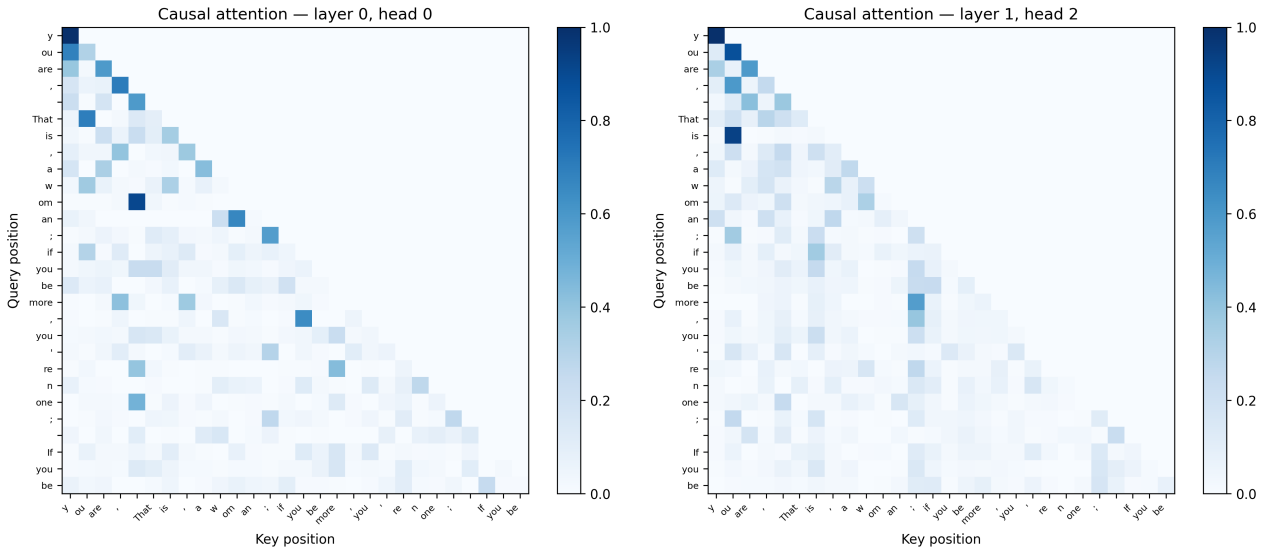

Figure 2: Causal attention on a fixed validation prefix. Left: layer 0, head 0. Right: layer 1, head 2.

3.3 How one head changes during training

Figure 3 holds the prefix, layer, and head fixed (layer 1, head 2) but swaps in weights from epochs 1, 5, 15, and 25. All four panels share one color scale, so brightness is comparable across time. Validation PPL at those checkpoints is listed in Table 1; it decreases monotonically as the maps evolve.

At epoch 1 the dominant feature is a tall vertical band on an early key (“That”): many later queries assign a large fraction of mass to the same token regardless of their own content. That is a simple strategy available to a nearly random head: attend to a salient early token. By epoch 5 the band is still there but the map is less uniform. By epoch 15 the column on “That” has weakened and several rows show sharp, interpretable peaks—the “is”–“ou” link is already strong. By epoch 25 the head no longer relies on one global column; instead, different queries distribute attention over several predecessors in a way that looks tuned to the specific prefix. The qualitative trajectory is therefore not monotonic sparsification: the map becomes more structured in the middle of training, then somewhat more diffuse again at convergence, while the loss metric improves throughout.

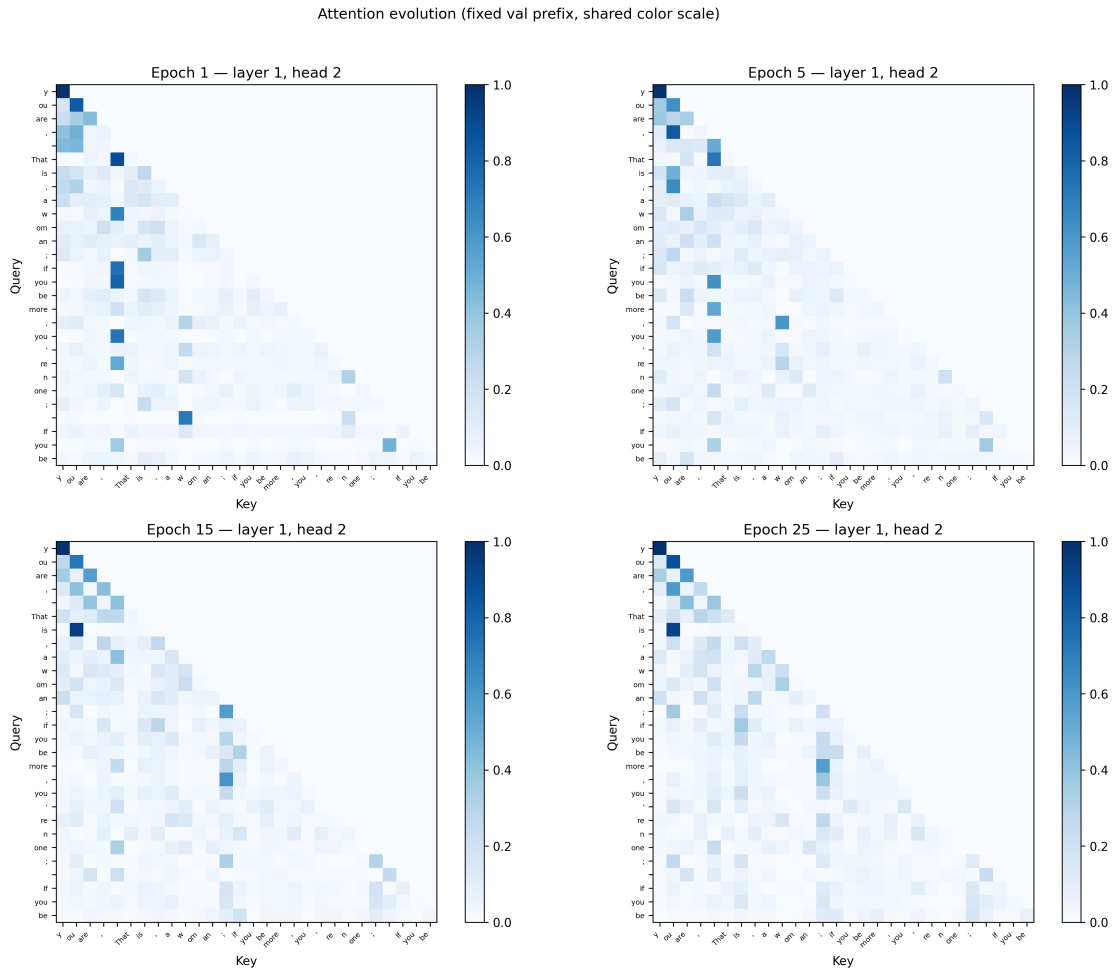

Figure 3: Layer 1, head 2 on the same prefix at epochs 1, 5, 15, 25 (shared color scale).

3.4 Entropy across heads and sensitivity to learning rate

The left panel of Figure 4 summarizes mean row entropy for each head in layer 1 on the same prefix used above. Entropy is $-\sum_j p_{ij} \log p_{ij}$ averaged over query rows i. High values mean the head spreads probability over many keys; low values mean each row is peaked on a small set.

Heads 3 and 7 have the lowest entropy (about 0.48 and 0.65 nats): they behave like sharp routers. Heads 0, 2, and 5 sit at the top of the range (about 2.08, 1.90, and 1.93) and behave more like wide integrators. Head 2, which Figures 2–3 visualize, falls in the upper middle of the distribution, which matches the observation that its maps are neither uniform nor one-hot on every row. The spread between the min and max head is a compact summary that "multi-head" here really is multi-behavior: heads are not interchangeable at this depth.

The right panel of Figure 4 shows three independent eight-epoch runs at 10^{-4}, 3×10^{-4}, and 10^{-3}. Final validation PPL is about 55.3, 46.0, and 41.4 respectively. A factor-of-ten increase in learning rate moves the endpoint close to the 25-epoch main run's PPL in only eight epochs, which underscores how strongly optimization speed depends on step size on this problem. It would be a mistake to read the chart as proof that 10^{-3} is always the right choice for 25 epochs or for other datasets: larger steps can diverge when clipping is removed or when the architecture changes. The main experiment still uses 3×10^{-4} for the long schedule as a conservative default. The full epoch-by-epoch traces show the same ordering at every step: 10^{-4} starts near 164 PPL at epoch 1 and descends slowly; 10^{-3} starts near 53 and flattens early.

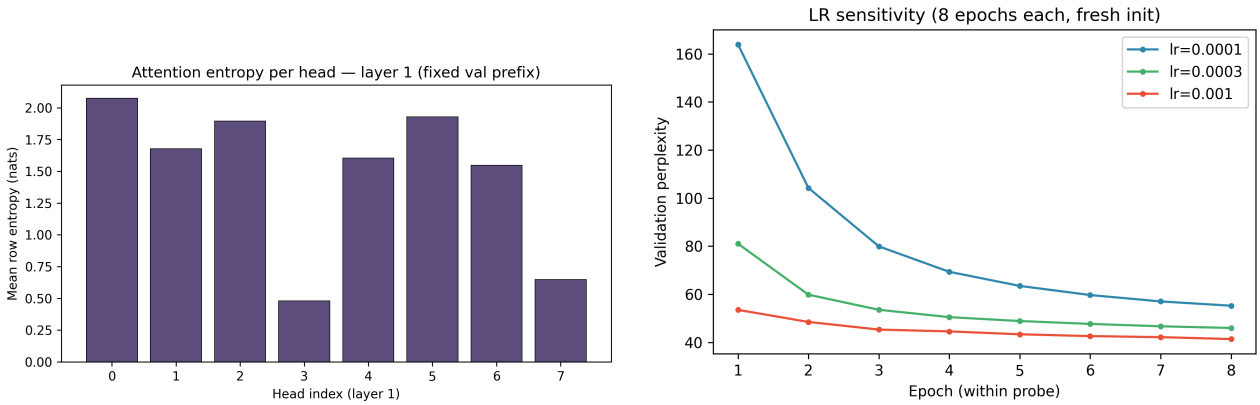

Figure 4: Left: mean attention row entropy per head in layer 1. Right: validation PPL for three learning rates (8 epochs each, fresh initialization).

3.5 Positional encodings and shorter context

Figure 5 (left) overlays validation perplexity for the main model truncated to 15 epochs and the no–positional-encoding model for the same 15 epochs. The sinusoidal curve lies below the no-PE curve at essentially every epoch; at epoch 15 the gap is about 0.45 PPL (43.13 versus 43.58). The effect is real but modest. One interpretation is that with only 64 supervised positions per window, and with Shakespeare's strong line breaks and capitalization, the model can partially infer order from token identity even when absolute position is not injected. A longer context or a less structured corpus would likely widen the gap.

The right panel trains a separate model with block length 33 and stride 16 for 12 epochs. Final PPL is 40.19, which is lower than the main run's final value, but the curves are not comparable as a pure "which model is better" statement: each prediction step conditions on half as many past tokens, so the per-step task is easier and the initial PPL is much lower (~ 63 versus ~ 83 at the first epoch). The practical point is compute: in this implementation, attention cost scales roughly with the square of sequence length among other terms, so halving the block length saves about a factor of four in that component at the price of throwing away tokens beyond the shorter horizon.

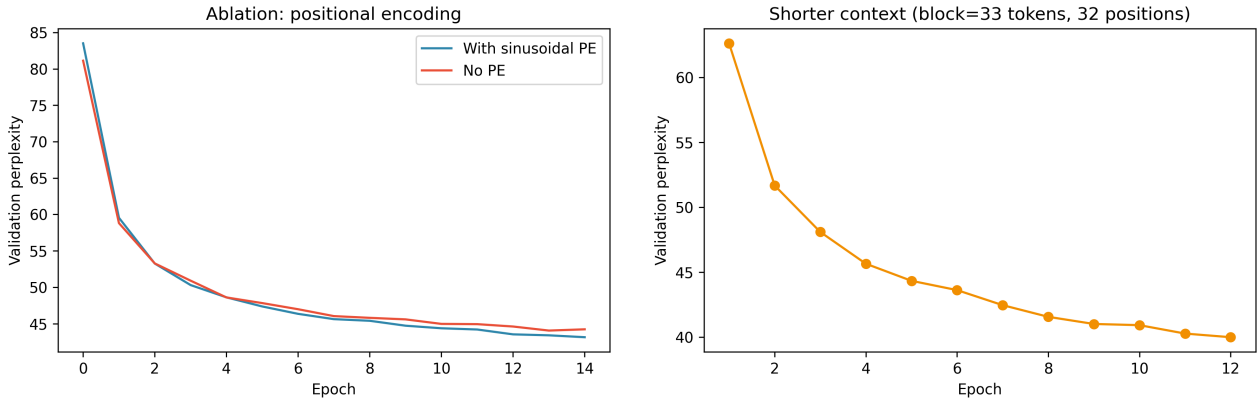

Figure 5: Left: positional encoding ablation (15 epochs). Right: shorter context (block 33, 12 epochs).

3.6 Sample generation

Sampling with temperature 0.85 from a short validation prompt produces comma-heavy fragments and archaic-looking spellings mixed with garbled invented subwords (for example "thereock", "prifft"). Coherence is limited, which is expected given model width, vocabulary size, and training budget.

4 Discussion

The training curves show a healthy optimization trajectory with mild train–val separation but no validation blowup. Attention visualizations make the causal constraint concrete and show that even a shallow network

can split roles across heads: one head in layer 0 emphasizes locality and punctuation, while a head in layer 1 forms sparse long-range links that track repeated words. The checkpoint sequence for layer 1, head 2 adds a temporal dimension: an early "anchor token" strategy is replaced by sharper structure mid-training, then by a more distributed pattern at convergence, all while perplexity decreases. That non-monotonic visual evolution is a useful reminder that interpretability plots are not the same as the loss; a head can look "messier" at the end while still representing a better function.

The entropy bars quantify head diversity at layer 1 and align with the qualitative contrast between peaked and diffuse maps. The learning-rate probe isolates step size as a dominant lever on an eight-epoch horizon, which helps explain why training recipes spend so much time on LR schedules and clipping. The PE ablation supports the standard claim that position information matters, while also showing that the benefit size depends on context length and data regularities. The short-context experiment is best read as a hardware-oriented counterpoint: smaller T buys quadratic savings.

This project stays on the supervised next-token objective. Methods that fine-tune with human feedback optimize a different signal—typically a scalar reward estimated from preferences—so the training loop and failure modes differ from cross-entropy on raw text. Scaling up depth, width, and context would change both perplexities and attention interpretations; the patterns described here should be treated as qualitative guides rather than universal laws.

5 AI Tool Usage Disclosure

I used interactive coding assistants for early notebook organization and plotting boilerplate. I wrote and debugged the model code, ran all training and experiments, exported figures `fig_01` through `fig_08`, and drafted this report, then checked every number against notebook output and `hw3_exp_*.txt`.